Start at 8:02 :)

ECON1050 Tutorial 3

Functions of One Variable

Week 4 - Semester 1 2026

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March 17, 2026

Today's Plan

- Recap: key ideas from Lecture 3 15 min
- Tutorial questions: Q1–Q5
- Further exercises: not done in tutorial...

Recap: What is a function?

Definition

A function of a real variable x with domain D is a rule that assigns a unique real number to each number x in D. As x varies over the whole domain, the set of all resulting values $f(x)$ is called the range of f.

Standard notation

$$\boxed{y = f(x)}$$

unique assign

$$\text{Domain} \xrightarrow{x} \boxed{f} \xrightarrow{f(x)} \text{Range}$$

A function maps the input set into an output set.

Vocabulary

- x: independent (exogenous) variable, or argument input
- y: dependent (endogenous) variable output
- **Domain**: all values that x can take $\{x\}$
- **Range**: the resulting values of y $\{f(x) : x \in D\}$

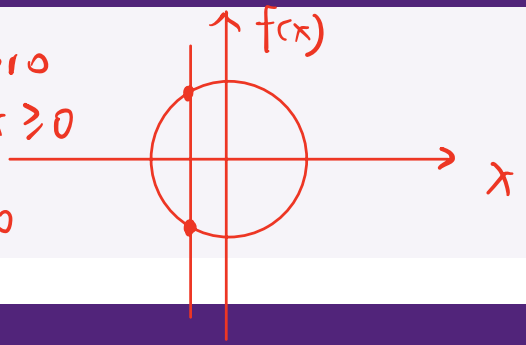
Recap: Graphs, domain, and monotonicity

Vertical line test

A graph represents a function if every vertical line intersects it at most once.

Typical domain restrictions

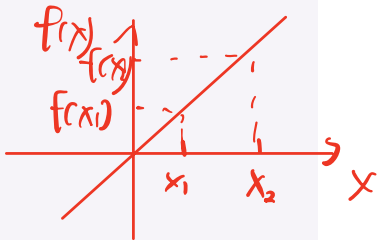
- Denominators cannot be zero. fraction $\frac{1}{x} \to$ can not be zero
- The argument of a square root must be nonnegative. $\sqrt{x} \Rightarrow x \geq 0$
- The argument of a logarithm must be positive. $\ln x \Rightarrow x > 0$



Increasing and decreasing functions

For $x_1 < x_2$ in an interval I:

$$f(x_1) \leq f(x_2) \iff \text{increasing}, \qquad f(x_1) \geq f(x_2) \iff \text{decreasing}.$$



The strict versions use $<$ and $>$ instead.

Recap: Key graph features

Common function types

- Linear: $f(x) = ax + b$
- Quadratic: $f(x) = ax^2 + bx + c$
- Hyperbolic: $f(x) = \dfrac{a}{x+b}$
- Cubic: $f(x) = ax^3 + bx^2 + cx + d$

What we look for in a graph

Intercepts, roots, asymptotes, and maximum or minimum points.

Coordinates

A point on a graph is written as an ordered pair (x, y).

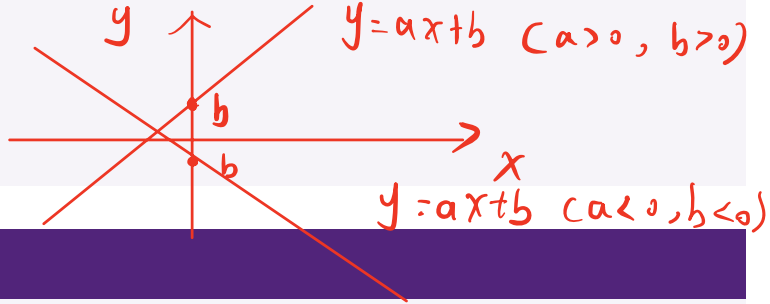
Recap: Linear functions and straight lines

General form

rate of change

$$y = ax + b$$

where a is the slope and b is the intercept on the y-axis.



Direction of the line

If $a > 0$, the function is increasing. If $a < 0$, the function is decreasing.

Useful formulas

Given two points, the slope is

$$\boxed{a = \frac{y_2 - y_1}{x_2 - x_1},} \qquad x_1 \neq x_2.$$

a, b ⇒ two equation
⇒ two point

If we know one point (x_1, y_1) and the slope a, then

$$\boxed{y - y_1 = a(x - x_1).}$$

$$\begin{cases} y = ax + b & ① \\ y_1 = ax_1 + b & ② \end{cases}$$

① − ② : $y - y_1 = a(x - x_1)$

Recap: Quadratic and polynomial functions

Quadratic functions

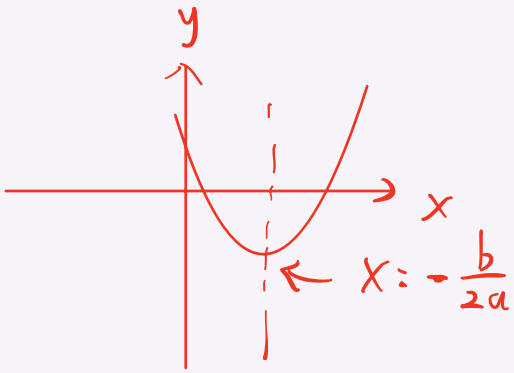
general form $y = ax^2 + bx + c$

A quadratic graph is a parabola. Its roots solve

$$\boxed{ax^2 + bx + c = 0,}$$

and the x-coordinate of the vertex is

$$x = -\frac{b}{2a}$$



Polynomial functions

A polynomial of degree n has the form

$$y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0. = \sum_{i=0}^{n} a_i x^i$$

A cubic function is a polynomial of degree 3.

Recap: Power functions and Cobb-Douglas

Power functions

The general form is

$$\boxed{y = Ax^a,}$$

where a is the exponent (or index) of x.

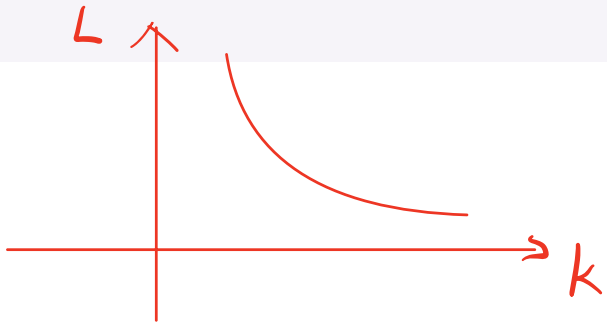
Cobb-Douglas production function

$$Q = AK^aL^b$$

Important

This describes output as a function of capital K and labour L.

Capital Labour

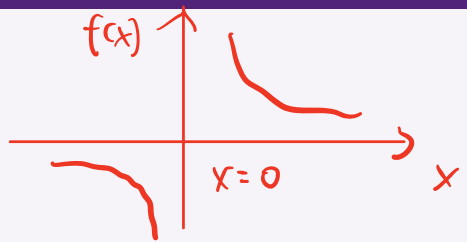


Recap: Hyperbolic and exponential functions

Hyperbolic functions

The simplest rational or hyperbolic function is

$$y = Ax^{-1} = \frac{A}{x}.$$



An **asymptote** is a line that the curve approaches as the variable tends to infinity.

Exponential functions

$$y = ba^x, \quad \boxed{f(t) = Aa^t} \qquad Aa^t$$

The variable appears in the exponent. Exponential functions are used for growth, decay, and compound interest.

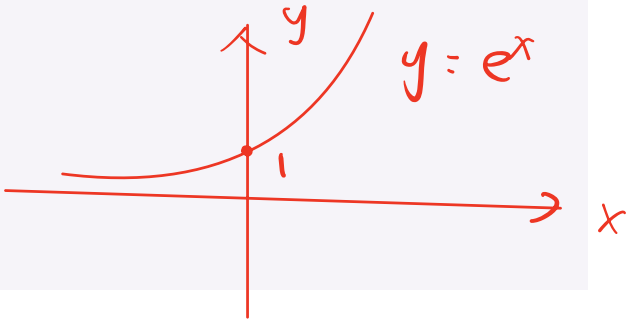
Recap: The natural exponential and logarithms

The natural exponential function

$$e \approx 2.718281828459, \qquad y = e^x.$$

Also,

$$e^{-x} = \frac{1}{e^x}.$$



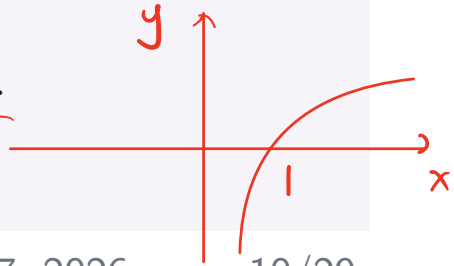
Logarithms

A logarithm is the exponent to which the base must be raised to produce a number. $y = \ln x$

Log rules

$$\ln(ab) = \ln a + \ln b, \qquad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \qquad \ln(x^c) = c \ln x.$$

$$e^{\ln x} = x, \qquad \ln(e^x) = x, \qquad \ln 1 = 0.$$



Recap: Transformations and new functions from old ones

Basic graph transformations

Starting from $y = f(x)$:

$$y = f(x) + c, \qquad y = f(x + c), \qquad y = cf(x), \qquad y = f(-x).$$

- $f(x) + c$: shift up or down
- $f(x + c)$: shift left or right
- $cf(x)$: vertical stretch (and reflection if $c < 0$)
- $f(-x)$: reflection about the y-axis

New functions from old ones

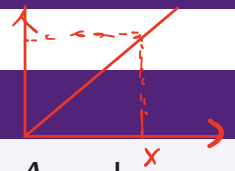
For functions f and g, we can form

time

$$F(x) = f(x) + g(x), \quad F(x) = f(x) - g(x), \quad F(x) = f(x)g(x), \quad F(x) = \frac{f(x)}{g(x)}, \quad F(x) = f(g(x)).$$

Recap: Inverse functions and economic interpretation

Inverse function



If f is one-to-one with domain A and range B, then it has an inverse function g with domain B and range A, defined by

$$g(y) = x \iff y = f(x), \qquad x \in A,\ y \in B.$$

Geometric characterization

If two functions are inverses of each other, then their graphs are symmetric about the line $y = x$.

Demand and supply

Algebraically, demand and supply may be written as

$$Q_d = D(P), \qquad Q_s = S(P),$$

so quantity is a function of price. Their inverses can be written as

$$P_d = d(Q), \qquad P_s = s(Q).$$

Demand Supply

$Q = D(P)$

$\Updownarrow$

$P = d(Q)$

Tutorial Q1

Question 1

Let

$$F(x) = 1 + \frac{4x}{x^2+4}.$$

(a) $F(0) = 1$

$F(-2) = 1 + \frac{4\cdot(-2)}{4+4} = 0$

$F(2) = 1 + \frac{4\cdot 2}{4+4} = 1 + 1 = 2$

$F(3) = 1 + \frac{4\cdot 3}{9+4} = \frac{25}{13}$

(b) $F(x) = 1 + \frac{4x}{x^2+4}$

$= 1 + \left[\frac{4}{x+\frac{4}{x}}\right] \to 0$

$x \to \infty$, $\frac{4}{x} \to 0$

$x + \frac{4}{x} = \infty + 0 = \infty$

$\lim_{x\to\infty} F(x) = 1$

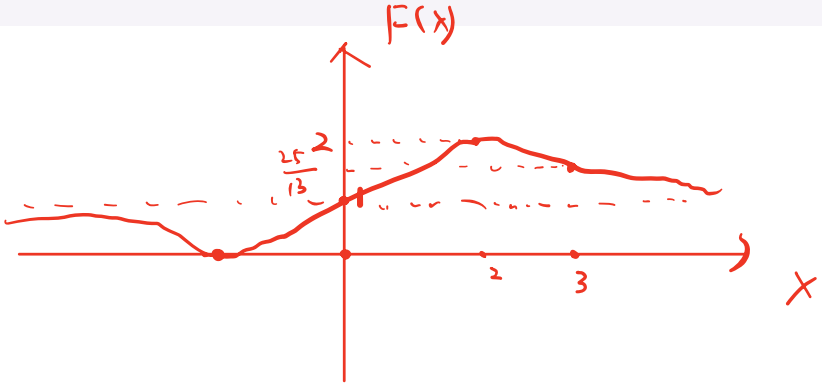
Similarly, $\lim_{x\to-\infty} F(x) = 1$

Verify!

(a) Evaluate $F(0)$, $F(-2)$, $F(2)$, and $F(3)$.

(b) Describe what happens to $F(x)$ when x becomes very large in magnitude.

(c) Sketch the graph using the key points from part (a) and the long-run behaviour from part (b).



Tutorial Q1: Function values and graph behaviour

Knowledge used

- Evaluate a function by direct substitution.
- Use algebra to study long-run behaviour and horizontal asymptotes.
- Sketch a graph from key points and asymptotic behaviour.

Solution

(a)

$$F(0) = 1, \qquad F(-2) = 0, \qquad F(2) = 2, \qquad F(3) = \frac{25}{13}.$$

(b)

$$F(x) = 1 + \frac{4x}{x^2+4} = 1 + \frac{4}{x + 4/x} \Rightarrow F(x) \to 1 \text{ as } x \to \pm\infty.$$

It approaches 1 from above when $x \to \infty$ and from below when $x \to -\infty$.

(c) Sketch through $(-2, 0)$, $(0, 1)$, $(2, 2)$, and $(3, 25/13)$ with horizontal asymptote $y = 1$.

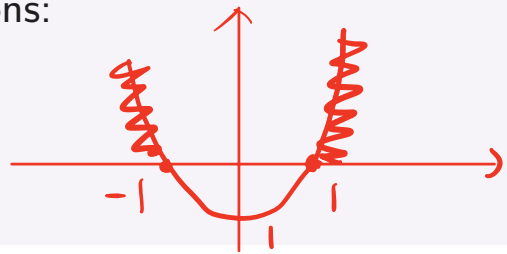
Tutorial Q2

Question 2

Find the domains of the following functions:

(a) $f(x) = \sqrt{x^2 - 1}$

(b) $g(x) = \dfrac{2x - 1}{x^2 - x}$



(a) $x^2 - 1 \geq 0 \Rightarrow \mathbb{R} \setminus (-1, 1)$ or $(-\infty, -1] \cup [1, \infty)$

(b) $x^2 - x \neq 0 \Rightarrow x(x-1) \neq 0 \Rightarrow x \neq 0$ and $x \neq 1$

$\Rightarrow \mathbb{R} \setminus \{0, 1\}$

Tutorial Q2: Domains of functions

Knowledge used

- For square roots, require the expression inside the root to be nonnegative.
- For rational functions, require the denominator to be nonzero.

Solution

(a)

$$x^2 - 1 \geq 0 \iff (x-1)(x+1) \geq 0.$$

Therefore

$$D_f = (-\infty, -1] \cup [1, \infty).$$

(b)

$$g(x) = \frac{2x-1}{x(x-1)}.$$

So we need $x \neq 0$ and $x \neq 1$, which gives $D_g = \mathbb{R} \setminus \{0, 1\} = (-\infty, 0) \cup (0, 1) \cup (1, \infty)$.

Question 3

Linear functions

(a) For the line $3x + 4y = 12$, find the slope and the y-intercept.

(b) Determine the equation of the straight line passing through $(-3, 5)$ and $(2, 7)$.

(a) $3x + 4y = 12 \Rightarrow$ general form of a linear function $y = ax + b$

$a \to$ slope, $b \to$ y-intercept

WTS

$4y = 12 - 3x$

$y = \frac{12-3x}{4} = 3 - \frac{3}{4}x \Rightarrow a = -\frac{3}{4}, \; b = 3$ ✓

(b) Given two points $(-3, 5)$ and $(2, 7) \Rightarrow$ find a linear function $\Rightarrow y = ax + b \Rightarrow a, b$

$$\begin{cases} 5 = -3a + b & ① \\ 7 = 2a + b & ② \end{cases} \Rightarrow$$

② − ①: $2 = 5a$

$a = \frac{2}{5}$ ③

plug ③ into ②: $7 = 2 \cdot \frac{2}{5} + b$

$b = 7 - \frac{4}{5} = \frac{31}{5}$ ✓

Tutorial Q3: Straight lines

Knowledge used

- Rewrite a linear equation into the form $y = ax + b$.
- Use the point-point formula to find the slope.
- Use the point-slope formula to recover the full equation.

Solution

(a)

$$3x + 4y = 12 \Rightarrow y = 3 - \frac{3}{4}x.$$

So the slope is $-\frac{3}{4}$ and the y-intercept is 3.

(b) The slope is $a = \frac{7-5}{2-(-3)} = \frac{2}{5}$. Using $(2, 7)$ in the point-slope formula,

$$y - 7 = \frac{2}{5}(x - 2) \Rightarrow y = \frac{2}{5}x + \frac{31}{5}.$$

Tutorial Q4

Question 4

Given

Demand function

$$Q_D = 150 - \frac{1}{2}P,$$

consumer

Supply function

$$Q_S = 20 + 2P,$$

producer

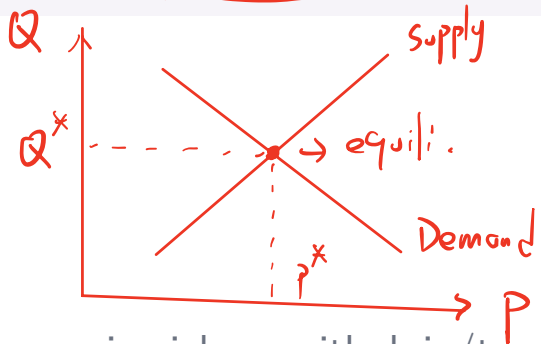
(a) Find the equilibrium price and quantity.

(b) Find the inverse demand and inverse supply functions.

(c) Suppose a tax of 2 is imposed on producers. Find the new equilibrium consumer price and quantity.

(d) Compare total revenue before and after the tax.

(a)



$$\begin{cases} Q_D = 150 - \frac{1}{2}P & ① \\ Q_S = 20 + 2P & ② \end{cases}$$

$\Rightarrow$ ① − ②: $130 - \frac{5}{2}P = 0$

$260 - 5P = 0$

$P^* = \frac{260}{5} = 52$ ③

plug ③ into ②: $Q^* = 20 + 2 \times 52$

$= 20 + 104 = 124$ ✓

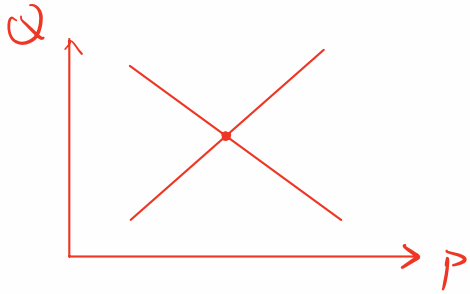
(b) Demand: $Q_D = 150 - \frac{1}{2}P \Rightarrow \frac{P}{2} = 150 - Q_D \Rightarrow P = 300 - 2Q_D$: inverse demand

Supply: $Q_S = 20 + 2P \Rightarrow 2P = Q_S - 20 \Rightarrow P = \frac{Q_S}{2} - 10$: inverse supply

(c) tax of 2 $\Rightarrow$ producers : P is the money earned

$\Rightarrow$ tax : producers earns $P-2$!

$Q_S = \boxed{20} + 2P \Rightarrow \begin{cases} Q_S = 20 + 2(P-2) = \boxed{16} + 2P & ① \\ Q_D = 150 - \frac{P}{2} & ② \end{cases}$



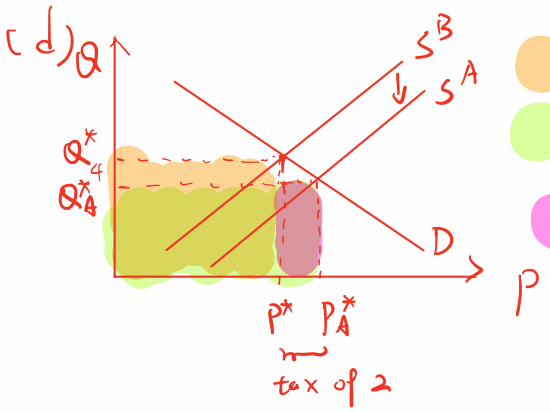
② - ①: $150 - 16 - \frac{P}{2} - 2P = 0$

$P^* = 53.6$ ③

plug ③ into ①: $Q^* = 16 + 2P^* = 16 + 107.2$

$= 123.2$ ✓

(d)



Before: $P^* \cdot Q^* = 52 \times 124 = 6448$

After: $P_A^* \cdot Q_A^* = 53.6 \times 123.2 = 6603.52$

tax Revenue: $2 \cdot Q_A^* = 2 \times 123.2 = 246.4$

Actual Revenue after tax: $P_A^* \cdot Q_A^* - 2Q_A^* = 6603.52 - 246.4$

–

Tutorial Q4: Equilibrium and inverse functions

Knowledge used

- Set demand equal to supply to find equilibrium.
- Rearrange a function to write price as a function of quantity.

Solution

(a) At equilibrium,

$$150 - \frac{1}{2}P^* = 20 + 2P^*.$$

Hence

$$130 = \frac{5}{2}P^* \Rightarrow P^* = 52, \quad \text{and} \quad Q^* = 150 - \frac{1}{2}(52) = 124.$$

(b) Inverse demand: $Q_D = 150 - \frac{1}{2}P \Rightarrow P = 300 - 2Q_D$.
Inverse supply: $Q_S = 20 + 2P \Rightarrow P = \frac{1}{2}Q_S - 10$.

Tutorial Q4: A producer tax

Knowledge used

- Adjust the supply equation to reflect the tax wedge.
- Distinguish between the consumer price and the producer price.

Solution

If producers pay a tax of 2, then $P_b = P - 2$, so the new supply relation becomes

$$Q_S^c = 20 + 2(P - 2) = 16 + 2P.$$

The new equilibrium satisfies

$$150 - \frac{1}{2}P_c^* = 16 + 2P_c^* \quad \Rightarrow \quad 134 = \frac{5}{2}P_c^* \quad \Rightarrow \quad P_c^* = 53.6.$$

The new equilibrium quantity is

$$Q_c^* = 150 - \frac{1}{2}(53.6) = 123.2.$$

Tutorial Q4: Revenue before and after the tax

Knowledge used

- Compute total revenue as price times quantity.
- Separate tax revenue from producer revenue after the tax.

Solution

$$TR^* = 52 \times 124 = \boxed{6448}, \qquad TR^{\text{with tax}} = 53.6 \times 123.2 = 6603.52.$$

$$T = 123.2 \times 2 = 246.4, \qquad \text{net producer revenue} = 6603.52 - 246.4 = 6357.12.$$

$$\text{Decrease in producer revenue} = 6448 - 6357.12 = 90.88.$$

Since $90.88 < 246.4$, consumers bear more than 60% of the tax.

Tutorial Q5

Question 5

Find the equation of the parabola passing through the three points

$$(1, -3), \qquad (0, -6), \qquad (3, 15).$$

Quadratic function

general form: $y = ax^2 + bx + c$

$a, b, c \Rightarrow$ Three points $\Rightarrow$ Three equations

$$\begin{cases} (1,-3): & -3 = a + b + c \quad ① \\ (0,-6): & -6 = 0 + 0 + c \quad ② \\ (3,15): & 15 = 9a + 3b + c \quad ③ \end{cases}$$

3 unknowns, 3 function

$\Rightarrow$ solve a system of 3 functions

$$\begin{cases} ① - ②: & \cancel{-3+6} \; 3 = a + b \quad ③ \\ ③ - ②: & \cancel{15+6} \; 21 = 9a + 3b \quad ④ \end{cases} \Rightarrow (a, b)$$

$3 \times ③$: $9 = 3a + 3b$ ⑤

④ − ⑤: $21 - 9 = 12 = 6a \Rightarrow \underline{a = 2}$ ⑥

plug ⑥ into ③ | ④: $3 = 2 + b \Rightarrow \underline{b = 1}$ ⑦

From ②, we know $\underline{c = -6}$

$y = 2x^2 + x - 6$ ✓ □

Tutorial Q5: Finding a quadratic function from three points

Knowledge used

- Start from the general quadratic form $y = ax^2 + bx + c$.
- Substitute each point to form a system of three equations.
- Solve for a, b, and c.

Solution

Let $y = ax^2 + bx + c$. Substituting the three points gives

$$a + b + c = -3, \qquad c = -6, \qquad 9a + 3b + c = 15.$$

So

$$a + b = 3, \qquad 3a + b = 7.$$

Subtracting these equations gives

$$2a = 4 \Rightarrow a = 2, \qquad b = 1, \qquad c = -6.$$

Therefore

$$y = 2x^2 + x - 6.$$

Further exercise: Q8

Knowledge used

- Combine a square-root restriction with rational-function restrictions.
- Use a sign diagram to determine where the expression is nonnegative.

Solution

For

$$g(x) = \sqrt{\frac{x-1}{(x-2)(x+3)}},$$

we need

$$\frac{x-1}{(x-2)(x+3)} \geq 0 \qquad \text{and} \qquad x \notin \{-3, 2\}.$$

A sign diagram gives

$$D_g = (-3, 1] \cup (2, \infty).$$

Further exercise: Q10

Knowledge used

- Form new functions from old ones: sum, difference, product, quotient, and composition.

Solution

Let

$$f(x) = x^3 - 2, \qquad g(x) = (1-x)x^2.$$

Then

$$f(x) + g(x) = x^2 - 2, \qquad f(x) - g(x) = 2x^3 - x^2 - 2.$$

$$f(x)g(x) = -x^6 + x^5 + 2x^3 - 2x^2.$$

$$\frac{f(x)}{g(x)} = \frac{x^3 - 2}{(1-x)x^2} = \frac{1}{1-x}\left(x - \frac{2}{x^2}\right).$$

For compositions,

$$g(1) = 0 \Rightarrow f(g(1)) = f(0) = -2, \qquad f(1) = -1 \Rightarrow g(f(1)) = g(-1) = 2.$$

Further exercise: Q11

Knowledge used

- Use logarithm rules to simplify expressions.
- Test identities carefully by rewriting both sides.

Solution

(a)

$$\ln\left(x^4 e^{-x}\right) = 4\ln x - x, \qquad \exp\left(\ln(x^2) - 2\ln y\right) = \frac{x^2}{y^2}.$$

(b) True:

$$\ln\frac{A}{B} + \ln\frac{B}{A} = 0, \qquad p\ln(\ln A) = \ln(\ln A)^p.$$

False:

$$p\ln(\ln A) = \ln(\ln A^p), \qquad \frac{\ln A}{\ln B + \ln C} = \ln\left(A(BC)^{-1}\right).$$

Further exercise: Q12

Knowledge used

- Rearrange a demand equation to write price as a function of demand.
- This is an inverse-function exercise.

Solution

(a)

$$D = \frac{32}{5} - \frac{3}{10}P \Rightarrow P = \frac{64}{3} - \frac{10}{3}D.$$

(b)

$$D = 120 - 5P \Rightarrow P = 24 - \frac{1}{5}D.$$

Key takeaways

- Check domains first: square roots, denominators, and logarithms all impose restrictions.
- For straight lines, know the slope-intercept form, the point-point formula, and the point-slope formula.
- Recognise the main functional forms: linear, quadratic, power, hyperbolic, exponential, logarithmic, and inverse functions.